

StyleMate: An AI-Powered Virtual Assistant for Personalized Clothing Selection Using 3D Body Modeling

Project ID – 25-26J-302

Project Proposal Report

Thivvijavearl T.S- IT22269270

Bachelor of Science (Hons) Degree in Information Technology Specializing in
Information Technology

Department of Information Technology

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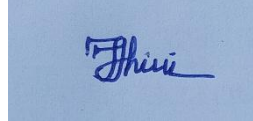
ABSTRACT

Occasion-Wise Outfit Recommendation System aims to reinvent personal styling decisions by addressing clothes selection issues in different events and even user interests through an AI-driven platform. Traditional fashion applications are prone to provide rigid, universal suggestions irrespective of personal context. To fill this gap, our system integrates Natural Language Processing (NLP), Machine Learning (ML), and Computer Vision to offer contextual and personalized outfit recommendations. Its core components are: (1) an NLP processor that understands user context in terms of occasion, mood, and style; (2) a machine learning-based algorithm for suggesting outfits based on learning from fashion databases and user behavior; (3) an interactive virtual mannequin for smooth out-of-the-box outfit previews; and (4) facial emotion detection for feeling user sentiment and dynamically adapting the suggestions. It is deployed as an easy-to-use web application with React for cross-platform compatibility, Firebase for secure and scalable storage, and Python-based machine learning libraries for intelligent decision-making. Performance is gauged by recommending accuracy, response time, and user satisfaction via surveys and prototype testing. Outside of everyday dress-up choices, the system has potential for e-commerce, online fashion buying, and virtual styling assistance. This work shows how AI, NLP, and Computer Vision can facilitate individualized fashion experiences, combining technology and life to create a more convenient, user-centric future for digital fashion advice.

Keywords

Occasion-Aware Outfit Recommendation, Natural Language Processing (NLP), Machine Learning (ML), Virtual Mannequin, Emotion Recognition, Personalized Fashion

Declaration

Name	Student ID	Signature
Thivvijavearl T.S	IT22269270	

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

.....
Signature of the Supervisor
Ms. Sanjeevi chandrasiri

.....
Date

.....
Signature of the Co-Supervisor
Ms.Karthiga Rajendran

.....
Date

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LIST OF ABBREVIATIONS

Key Words	Meaning
GCP	Google Cloud Platform
DB	Database
UI	User Interface
API	Application Programming Interface

Table 1 List of Abbreviations

INTRODUCTION

One of the most common dilemmas that one encounters in life is what to wear to an event. Fashion choice can be a stressful and frustrating endeavor, especially with the growing variety of styles, colors, and accessories that one is exposed to. Conventional methods of selecting an outfit greatly rely on one's personal judgment or general fashion advice, which are not typically personalized and do not include major factors such as body shape, mood, complexion, and event type.

With the advent of Artificial Intelligence (AI) and Machine Learning (ML), it is possible now to have intelligent systems that deliver user-specific, context-based apparel suggestions. Utilizing Natural Language Processing (NLP), user preferences and event descriptions can be interpreted, while Computer Vision (CV) and emotional detection add to personalization with body type analysis and user response feedback.

Current advancements in AI-powered recommendation platforms demonstrate significant improvements in precision and user satisfaction. For instance, NLP techniques can convert text input like "wedding," "business meeting," or "casual get-together," and map them to suitable classes of attire. Similarly, emotion recognition models, developed on deep learning and OpenCV, can scan live user responses to fine-tune suggestions.

The proposed system, StyleMate: An AI-Based Virtual Fashion Assistant, aims to shatter the limitations of existing fashion apps through the combination of occasion-based recommendation, virtual mannequin preview, and sentiment-based personalization. The system will be built as a web application for convenient access and utilization, employing React for cross-platform deployment, Firebase for handling large volumes of data, and Python (with TensorFlow and spacy frameworks) for machine learning and NLP processing. Through dynamic, adaptive, and personalized fashion suggestions for outfits, StyleMate assists people in making knowledge-based fashion decisions appropriate to their personality and event context.

1.2 Background and Literature survey

Artificial Intelligence (AI), Natural Language Processing (NLP), and Computer Vision (CV) are all used to a large extent to make fashion recommendations more personalized. These technologies allow systems to recommend clothing by accomplishing user requirements, occasions, and visual traits.

- **AI-based Fashion Recommendation:** AI algorithms like collaborative filtering, decision trees, and deep learning (CNNs and RNNs) can accurately identify suitable apparel. Li et al. (2021), for example, found that AI-based fashion systems increased user satisfaction by over 40% compared to basic recommendations.
- **Natural Language Processing in Styling:** NLP models such as spaCy, BERT, and GPT can understand user inputs such as "wedding," "formal event," or "casual dinner." Kim et al. (2022) showed how NLP can effectively map these inputs to the respective clothing categories and increase recommendation relevance.
- **Globalization and Fashion:** Virtual mannequins and try-on systems are developed through TensorFlow, OpenCV, and GANs methods. Huang et al. (2020) developed a system that enables users to preview clothes in a realistic manner and make appropriate choices.
- **Emotion Recognition in Retail:** Emotion recognition through facial expressions can authenticate users' satisfaction in real time. Smith et al. (2023) verified that the use of emotions in recommendations gives suggestions more sensitive and reactive to user feedback.

1.1 Research Gap

1.3 Research Gap

As much as AI-powered fashion recommendation systems have been encouraging with results, current models tend to fall short of providing context-aware and individualized outfit suggestions. Generic fashion advice tends to be provided by most systems without consideration for basic factors like occasion, emotional state of the user, body shape, and fashion sense. Furthermore, real-time feedback integration through recognition of emotions is rarely practiced. Most of the current platforms provide static suggestions or employ prior purchase history only, leading to poor suggestions and choice overload. Since most of the current systems are never optimized to take NLP, virtual try-ons, and emotion-based personalization together into one single platform, the users find inefficiency in choosing the right clothes.

The objective of this study is to fill these gaps by:

- Developing an AI-powered suggestion system considering occasion, style, mood, and history to suggest suitable outfits.
- Using a virtual mannequin to display real-time previews and accurate visualizations of suggested outfits.
- Using facial emotion detection to dynamically adapt based on feedback from the user, maximizing satisfaction.
- Incorporating a simple-to-use web app for seamless interaction, making personalized fashion tips accessible to every user.

By bridging these gaps, this project aims to enhance personalization in fashion, reduce anxiety in making decisions, and offer adaptive, occasion-dependent outfit suggestions.

COMPARISON TO OLD MODELS

Old Model	New Model
Most fashion apps give generic recommendations based on popularity or limited user preferences. They rarely consider occasion, mood, body type, or real-time feedback, resulting in less accurate suggestions.	The Occasion-Wise Outfit Recommendation system uses AI, NLP, and emotion recognition to provide context-aware, personalized outfit suggestions. It includes a virtual mannequin for real-time previews and adapts recommendations based on user reactions.

Table 1 compare to old model

1.2 Research Problem

The primary research problem addressed in this study is that users are struggling to select the right outfits for an event, which they tend to find causing decision fatigue, poor dressing choice, and loss of confidence. Traditional fashion guidance systems do not work effectively, as they provide vague recommendations based on popularity or historic buy history and barely take into account factors such as body shape, mood, individual taste, or latest feedback.

The absence of an occasion-aware, AI-based fashion assistant that can integrate affect recognition, user preferences, and virtual try-on is the biggest hindrance. Current solutions rely on static data or clear features that fail to deliver a seamless, personalized styling experience.

This work aims to answer the following research questions:

1. How can machine learning and NLP be utilized to garner user context, occasion, and preference data in order to provide individualized outfit suggestions?
2. How can virtual mannequins and computer vision be utilized to deliver real-time, realistic previews of clothes on users?
3. How can facial emotion recognition be integrated to dynamically generate recommendations and improve user satisfaction and trust in the choice of outfits?

Through the answers to these issues, the aim of this project is to develop the Occasion-Wise Outfit Recommendation system, allowing users to receive real-time, personalized, and context-sensitive outfit guidance, improving decision-making and overall fashion experience.

2 OBJECTIVES

2.1 Main Objectives

The primary objective of this project is to develop an Occasion-Based Outfit Recommendation system that leverages Artificial Intelligence (AI), Natural Language Processing (NLP), Machine Learning (ML), and Computer Vision (CV) to provide personalized outfit recommendations for different occasions. Traditional fashion apps provide generic recommendations, which can lead to decision fatigue, mismatched outfits, and low confidence. Users find it difficult to decide on apparel to suit their body type, mood, personal taste, and occasion. To address these issues, the system suggested here will take user inputs, past selections, and real-time feedback to suggest suitable apparel combinations. Using AI-based models and virtual mannequins, the system suggested here will provide realistic outfit previews and dynamically alter suggestions based on user feedback. Additionally, a user-friendly web app will make personal styling easily accessible, interactive, and convenient. Finally, this project aims to increase user confidence, remove decision-making anxiety, and improve personalization in fashion recommendations.

2.2 Specific Objectives

1. Create a web application for real-time outfit suggestions and previews.
2. Implement NLP models for capturing user context, style preferences, and occasion details.
3. Train machine learning models for outfit suggestions based on fashion datasets and user history.
4. Incorporate virtual mannequins for realistic, interactive outfit visualization.
5. Use facial emotion recognition to personalize recommendations dynamically based on user reactions.
6. Create a voice-enabled or interactive chatbot for accessory suggestions and styling tips.
7. Ensure secure storage and management of user tastes and interaction history.
8. Ensure cross-platform compatibility and accessibility features for a broad user base.
9. Evaluate the system's performance through user testing and feedback analysis.
10. Promote adoption by highlighting added personalization and convenience over existing fashion apps.

3 METHODOLOGY

The Occasion-Wise Outfit Recommendation system design follows a structured framework to deliver efficient, personalized, and responsive outfit suggestions. It begins with data collection where fashion datasets will be fetched from web repositories (e.g., Deep Fashion, Fashion-MNIST, VITON) alongside user input in the form of body measurements, style preference, and occasion information. Past user interaction history and feedback will also be collected for enhancing the accuracy and relativity level of the recommendations.

Once data is collected, it will be preprocessed to standardize and validate it. Preprocessing includes cleaning missing or inconsistent data values, normalization of clothing attributes, and transforming textual inputs into encodings for NLP models. Clean and well-structured data will improve the precision of machine learning models, which will be trained to hear user preferences, fashion trends, and occasion context and suggest suitable outfits. The system will employ supervised learning algorithms, i.e., decision trees, neural networks, and collaborative filtering techniques, to generate recommendations dynamically based on user data and feedback. The models will be updated in real time with new data coming in, hence delivering adaptive and context-sensitive suggestions.

To enhance personalization, the system will integrate virtual mannequins via Computer Vision and 3D modeling, allowing customers to see clothing realistically on a digital replica of their body. Facial emotion detection will be applied to analyze user emotions in real-time, dynamically adjusting recommendations for higher satisfaction. An interactive voice-enabled chatbot will also provide advice on accessory compatibility and styling tips, enhancing accessibility and user engagement.

The web app based on React will possess a user-friendly interface, real-time outfit preview, personalization-based recommendations, support for notifications, and multilingualism. The backend would be developed in Python using TensorFlow and spacy as the libraries for ML and NLP and using Firebase as the secure and scalable data storage.

By the integration of AI-based recommendation models, NLP-based user understanding, virtual try-ons, and emotional detection, the Occasion-Wise Outfit Recommendation system aims to allow users to make decision-aware and personalized fashion choices with reduced decision fatigue and enhanced styling experience overall.

Project Methodology

Occasion-Wise Outfit Recommendation System

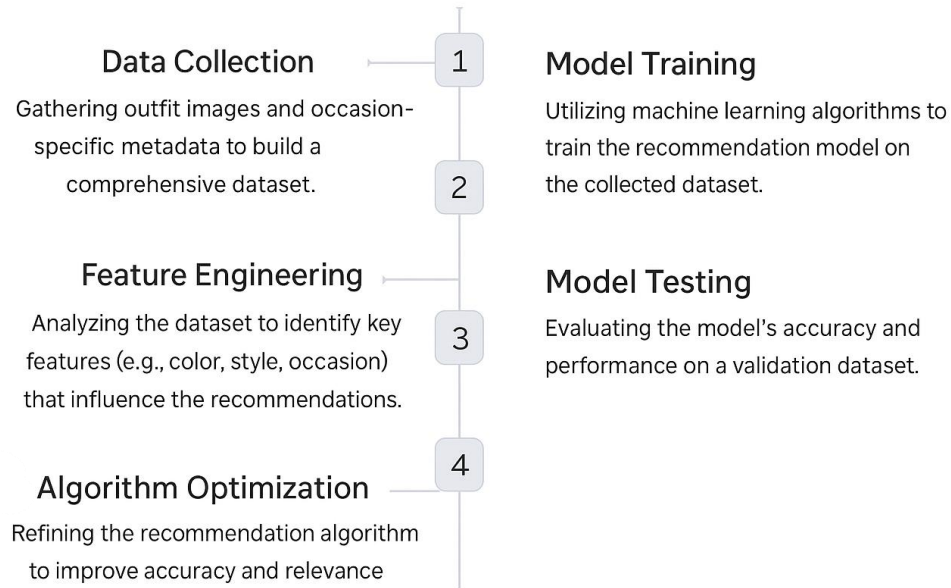


Figure:1methodology

Occasion-Wise Outfit Recommendation system adopts a three-layer architecture to ensure seamless integration of AI models, virtual mannequins, and intuitive interfaces. The three layers are:

1. **User Interface (UI) Layer:** Provides users with an interactive web app for real-time outfit recommendations and previews.
2. **Controller Layer:** Acts as an interface between the UI and backend, processing user inputs, occasion details, and feedback to generate personalized outfit recommendations.
3. **Backend Services and Database Layer:** It is concerned with data storage, machine learning and NLP model execution, and analytics to facilitate intelligent recommendations.

3.1 User Interface (UI) Layer

The UI Layer is the front end of the system that presents the users with a consistent and interactive experience. Some of its key features include:

- **Dashboard:** Displays real-time outfit suggestions, occasion details, and personalized styling tips.
- **Virtual Mannequin Preview:** Allows users to preview shortlisted outfits realistically on a 3D virtual model of their body.
- **Recommendation Insights:** Provides actionable recommendations on suitable apparel combinations by occasion, style, and mood.
- **Feedback System:** Allows users to rate outfit suggestions and provide feedback to improve the recommendation engine.

3.2 Controller Layer

Controller Layer processes user inputs, virtual mannequin interactions, and feedback, communicating with the backend to send results to the UI. Primary functions include:

- **Data Preprocessing:** Sanitizes and formats user inputs, fashion dataset features, and occasion text descriptions for analysis.
- **Outfit Recommendation:** Utilizes machine learning and NLP models to predict suitable clothing combinations for specified occasions.
- **Feedback Management:** Collects user ratings and comments for model performance enhancement and personalization.

3.3 Backend Services and Database Layer

The Backend Layer is accountable for storing data, running AI models, and performing analytics. The key components include:

- **Machine Learning Models:** Neural networks, collaborative filtering, and NLP-based classifiers algorithms anticipate personalized outfit suggestions.
- **Data Storage:** Firebase or MongoDB stores user profiles, fashion datasets, preferences, and model updates.

- **Feedback Collection:** Ratings and user interactions are collected and used for retraining models for continuous improvement.
- **Analytics Engine:** Examines historical and real-time data in order to tune recommendations and reveal trends in user preferences.

Data Storage: Firebase Cloud Storage is used to store sensor data, user profiles, and model updates.

Feedback Collection: User feedback is collected and used to retrain machine learning models for continuous improvement.

4 Gantt chart



Gantt Chart

25-26J-302 | StyleMate Web App

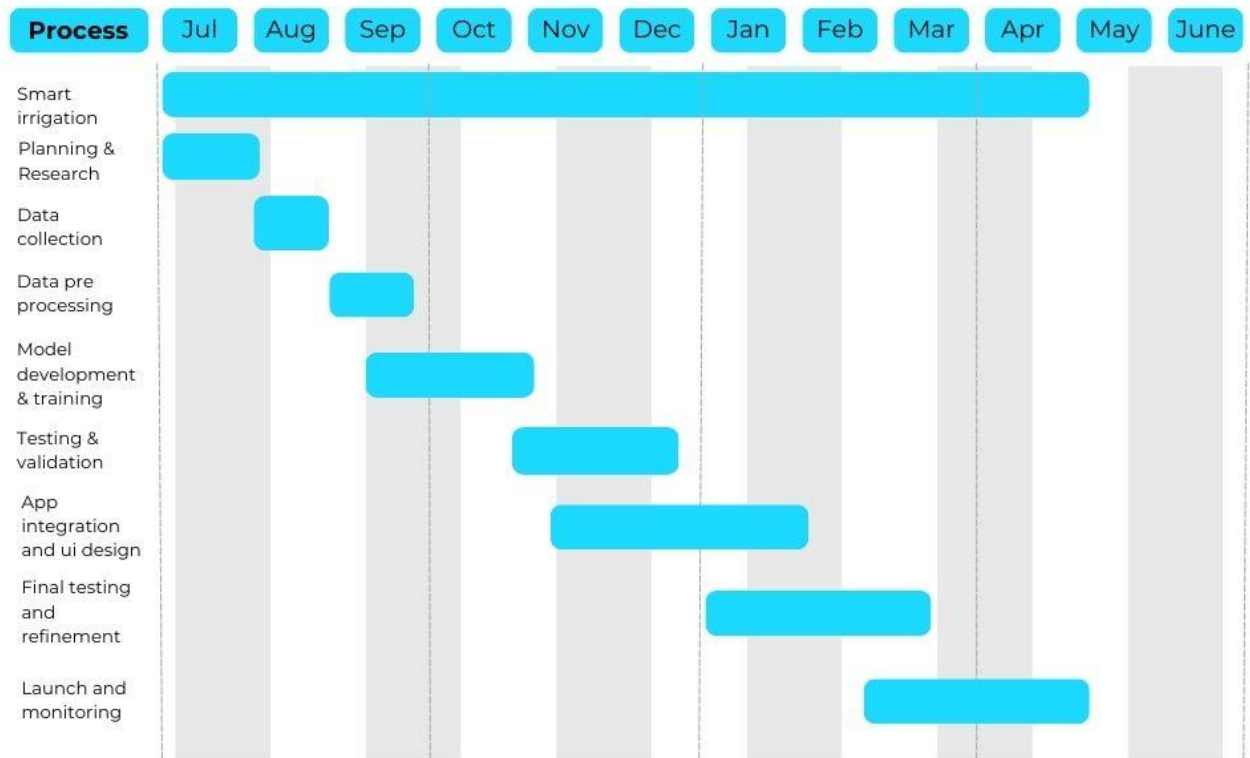


Figure:Error! Use the Home tab to apply 0 to the text that you want to appear here.:2 Gantt chart

5 Project requirements

5.1 Functional Requirements

1. User Authentication & Management: Integrate secure user login and registration using Firebase Authentication.
2. Profile & Preference Management: Allow users to input body measurements, style, and occasion data.
3. Outfit Recommendation Engine: Offer customized outfit suggestions using machine learning and NLP algorithms.
4. Virtual Mannequin Preview: Show suggested outfits on a 3D virtual mannequin for realistic visualization.
5. Feedback System: Allow users to rate suggestions and provide comments for continuous improvement.
6. Multilingual Support: Ensure the application is accessible to users across different languages.
7. Interactive Chatbot: Provide accessory pairing and styling recommendations through a voice chatbot.

5.2 Nonfunctional Requirements

1. Performance: The system should offer outfit recommendations within 3–5 seconds.
2. Scalability: The system should be able to cope with more users and datasets without impacting performance.
3. Data Security: User data, including preferences and body measurements, should be encrypted and safely stored using Firebase security rules.
4. Availability: The system must maintain 99.9% uptime for seamless service.
5. Usability: The web app must provide a simple-to-use interface suitable for users with varying levels of digital literacy.

5.3 Technology Selection

Machine Learning & NLP: Python with TensorFlow, Scikit-learn, and spaCy for recommendation models and text processing.

Computer Vision: OpenCV and GANs for virtual mannequin creation and clothing visualization.

Web app: web application.

Database & Cloud Storage: Firebase for storing real-time data, user profiles, and model updates.

Voice & Chatbot: voice transformers and speech recognition APIs for interactive guidance.

6 System diagram

StyleMate: AI-Powered Virtual Assistant for Personalize Clothing Selection Using 3D Body Modeling

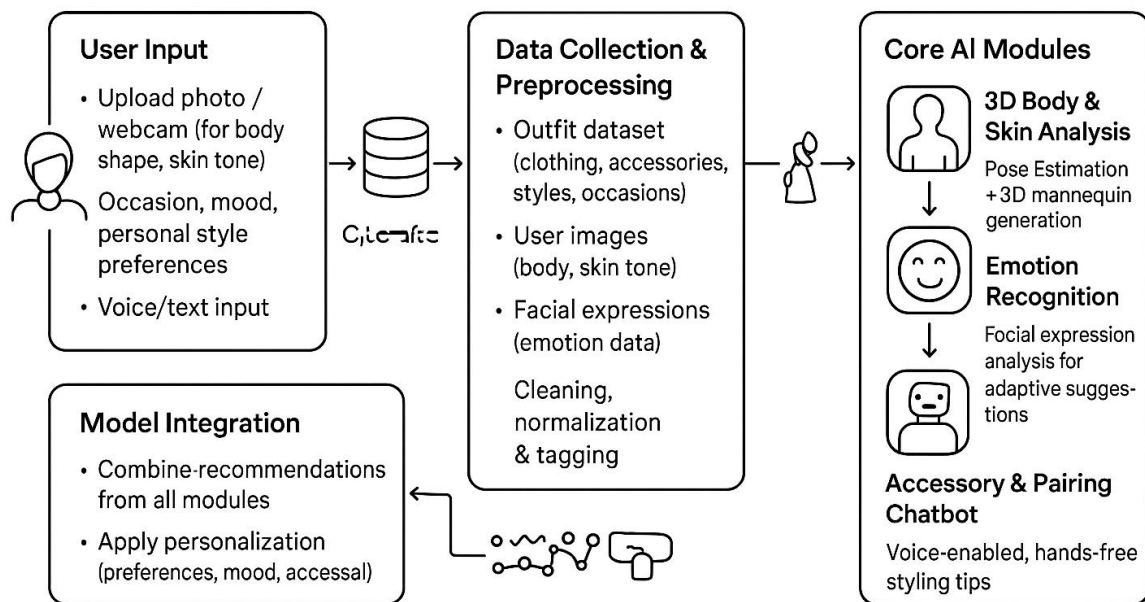


Figure:3System diagram

7 CONCLUSION

The project is conceptualized to enhance personalized fashion experiences, outfit choice efficiency, and customer satisfaction through the use of AI, NLP, Computer Vision, and data-driven decision-making. The StyleMate system will provide users with real-time outfit recommendations, virtual try-on, and predictive styling analytics for effectively optimizing fashion choices. With occasion-based input and AI-driven recommendations, the system will offer the users an outfit that matches their preference, body type, and occasion requirement, omitting decision fatigue and improving style confidence altogether.

Furthermore, with the utilization of a user-friendly web app and interactive chatbot, users will be able to navigate styling options efficiently, receive accessory suggestions, and try on outfits virtually. The scalability and multilingual nature of the system ensure usability for diverse users.

The solution has the potential to transform traditional fashion recommendation methods by making intelligent, personalized, and context-aware outfit suggestions possible. Potential future extensions involve integration with AR/VR try-on sessions, social media sharing, and AI-driven trend prediction. By addressing some of the key challenges of fashion personalization, the project provides a long-term, technology-driven solution for style management, increasing user satisfaction and engagement for online fashion.

The success of this project will lead to advancements in AI-driven fashion tech innovations and open doors for further innovation in personalized styling products.

8 BUDGET AND BUDGET JUSTIFICATION

Item	Description	Cost Estimation
Hosting & Cloud	Basic cloud server / GPU rental for 6 months	30,000 .00
Domain & Web Hosting	Domain (stylemate.lk) + basic web hosting	10,000.00

Publication	Conference/journal fees + printing/binding	12,500 .00
Deployment	Cloud deployment of prototype (3–4 months)	15,000.00
Marketing (Optional)	Posters, brochures, small-scale promotions	20,000.00
	Total Estimated	87,500.00

Table 2 Budget and Budget Justification

9 COMMERCIALIZATION

9.1 Market Opportunity

The AI-powered fashion recommendation system market is expanding rapidly, targeting individual users, e-commerce sites, and fashion retailers. Accurate and personalized outfit suggestions can be used to drive user engagement, online sales, and the overall fashion shopping experience..

9.2 Revenue Model

B2C Freemium Model: Basic free app with premium paid features such as advanced recommendations and virtual try-on.

B2B Licensing: Licensing AI styling solutions to fashion retailers and e-commerce sites.

Affiliate Marketing: Earning commissions for linking recommended outfits to online stores.

Advertising: In-app advertisements for fashion labels and accessories.

Data Insights: Anonymous trend and style preference information sold to fashion brands.

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9 APPENDICES

Technology Stack:

Frontend: React (web application)

Backend: Python (Machine Learning & NLP)

Database & Cloud Storage: Firebase

System Diagrams:

User Interaction Flow

Machine Learning Recommendation Pipeline

Virtual Mannequin Rendering & Outfit Visualization